

# Trust & Safety Moderation Action Prediction using Logistic Regression

## Project Overview

This project develops a Logistic Regression model to predict moderation actions using behavioral signals commonly observed in Trust & Safety operations.

Unlike purely rule-based systems, this dataset introduces uncertainty and overlapping behavioral patterns to better simulate real-world moderation environments.

Predicted moderation actions include:

- Allow
- Warn
- Restrict
- Ban

---

## Dataset Description

The dataset contains 2,000 moderation records designed to approximate realistic Trust & Safety workflows.

### Features

| Feature             | Description                                            |
|---------------------|--------------------------------------------------------|
| User ID             | Unique user identifier                                 |
| Violative           | Indicates whether the current activity violates policy |
| Frequency           | Number of violations committed                         |
| Timestamp (minutes) | Time between incidents                                 |
| Previous Violations | Historical violation count                             |
| Severity            | Severity level of the incident                         |
| Action              | Moderation action taken (Target Variable)              |

---

## Problem Statement

Predict the appropriate moderation action based on behavioral and historical user signals.

This problem is formulated as a multiclass classification task.

---

## Data Preprocessing

The following preprocessing steps were performed:

- Removed `User ID`
  - Encoded categorical variables
  - Scaled numerical features using `StandardScaler`
  - Applied stratified train-test split
- 

## Logistic Regression Model

```
model = LogisticRegression(max_iter=20000)
```

---

## Model Evaluation

The Logistic Regression model was evaluated using multiple classification metrics.

### Performance Metrics

| Metric            | Score |
|-------------------|-------|
| Accuracy          | 68%   |
| Macro Precision   | 63%   |
| Macro Recall      | 59%   |
| Macro F1-Score    | 59%   |
| Weighted F1-Score | 64%   |

### Classification Report

| Action   | Precision | Recall | F1-Score | Support |
|----------|-----------|--------|----------|---------|
| Allow    | 0.72      | 0.97   | 0.82     | 172     |
| Warn     | 0.60      | 0.38   | 0.47     | 92      |
| Restrict | 0.45      | 0.28   | 0.35     | 71      |

| Action | Precision | Recall | F1-Score | Support |
|--------|-----------|--------|----------|---------|
| Ban    | 0.74      | 0.74   | 0.74     | 65      |

## Interpretation

- The model performs exceptionally well in identifying **Allow** actions, achieving a recall of **97%**.
- The model demonstrates strong performance for **Ban** decisions, with both precision and recall reaching **74%**.
- Performance on **Warn** and **Restrict** classes is comparatively lower, indicating overlapping behavioral patterns and ambiguity in moderation decisions.
- The overall accuracy of **68%** suggests that the dataset contains realistic uncertainty, making it a more representative approximation of real-world Trust & Safety moderation environments.

## Key Insight

The moderate performance on intermediate enforcement actions (**Warn** and **Restrict**) reflects the complexity of moderation systems, where similar behavioral signals may lead to different enforcement outcomes depending on context and historical user behavior.

- Moderation decisions are influenced by multiple behavioral signals.
- No single feature completely determines the enforcement action.
- The dataset contains ambiguity and overlapping classes, resulting in more realistic performance.

## Dataset Characteristics

### Action Distribution

| Action   | Approximate Percentage |
|----------|------------------------|
| Allow    | 43%                    |
| Warn     | 23%                    |
| Restrict | 18%                    |
| Ban      | 16%                    |

This distribution reflects typical platform moderation scenarios where most users remain compliant.

## Strengths

- More realistic than rule-based datasets.
- Includes behavioral ambiguity.
- Produces believable model performance.
- Suitable for demonstrating model evaluation and error analysis.

---

## Limitations

The dataset is still synthetic and does not include:

- Actual user-generated text
- Human disagreement
- Contextual understanding
- Sarcasm
- Appeals and policy exceptions

---

## Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib

---

## Future Improvements

- Incorporate NLP-based content signals.
- Evaluate on human-annotated moderation datasets.
- Compare multiple machine learning algorithms.
- Build a moderation dashboard using Streamlit.

---

## Conclusion

This project demonstrates the application of Logistic Regression in Trust & Safety moderation and provides a more realistic approximation of human moderation behavior by introducing uncertainty and overlapping decision boundaries.